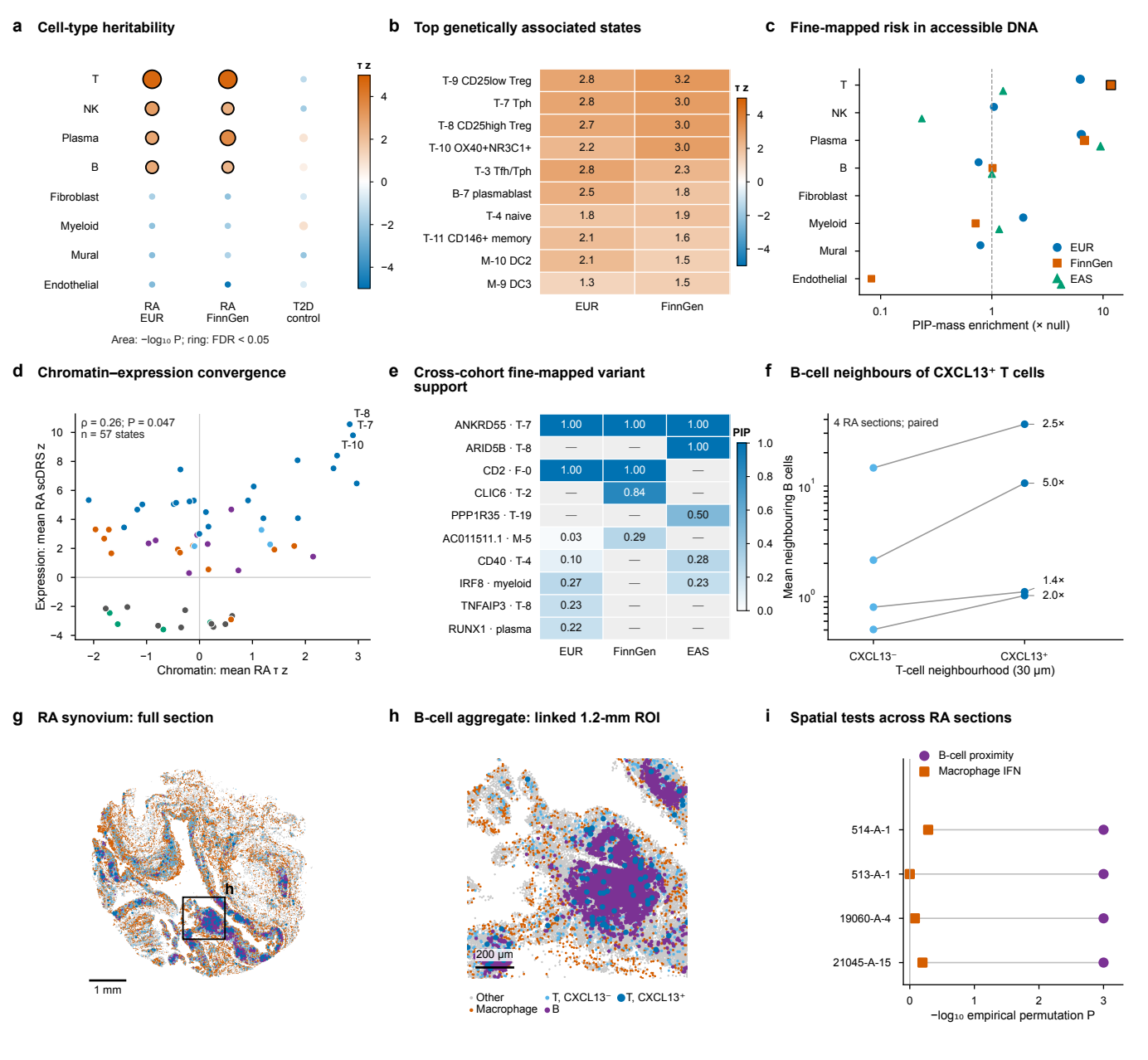

a
Cell-type heritability
T
NK
Plasma
B
Fibroblast
Myeloid
Mural
Endothelial
RA EUR
RA FinnGen
T2D control
τ z
4
2
0
−2
−4
Area: −log10 P; ring: FDR < 0.05
b
Top genetically associated states
T-9 CD25low Treg 2.8 3.2
T-7 Tph 2.8 3.0
T-8 CD25high Treg 2.7 3.0
T-10 OX40+NR3C1+ 2.2 3.0
T-3 Tfh/Tph 2.8 2.3
B-7 plasmablast 2.5 1.8
T-4 naive 1.8 1.9
T-11 CD146+ memory 2.1 1.6
M-10 DC2 2.1 1.5
M-9 DC3 1.3 1.5
EUR
FinnGen
τ z
4
2
0
−2
−4
c
Fine-mapped risk in accessible DNA
T
NK
Plasma
B
Fibroblast
Myeloid
Mural
Endothelial
EUR
FinnGen
EAS
0.1
1
10
PIP-mass enrichment (× null)
d
Chromatin–expression convergence
ρ = 0.26; P = 0.047
n = 57 states
T-8
T-7
T-10
Expression: mean RA scDRS z
Chromatin: mean RA τ z
e
Cross-cohort fine-mapped variant support
ANKRD55 · T-7 1.00 1.00 1.00
ARID5B · T-8 — — 1.00
CD2 · F-0 1.00 1.00 —
CLIC6 · T-2 — 0.84 —
PPP1R35 · T-19 — — 0.50
AC011511.1 · M-5 0.03 0.29 —
CD40 · T-4 0.10 — 0.28
IRF8 · myeloid 0.27 — 0.23
TNFAIP3 · T-8 0.23 — —
RUNX1 · plasma 0.22 — —
EUR
FinnGen
EAS
PIP
1.0
0.8
0.6
0.4
0.2
0.0
f
B-cell neighbours of CXCL13+ T cells
4 RA sections; paired
2.5×
5.0×
1.4×
2.0×
Mean neighbouring B cells
CXCL13−
CXCL13+
T-cell neighbourhood (30 µm)
g
RA synovium: full section
h
1 mm
h
B-cell aggregate: linked 1.2-mm ROI
200 µm
Other
T, CXCL13−
T, CXCL13+
Macrophage
B
i
Spatial tests across RA sections
B-cell proximity
Macrophage IFN
514-A-1
513-A-1
19060-A-4
21045-A-15
0
1
2
3
−log10 empirical permutation P